

Employment

Product Engineer Intern **Isomer AI** **May 2025 – May 2026**

- Ship 10+ production AI features end-to-end: multi-model LLM pipelines (Claude, GPT-4o, Gemini), Temporal.io agentic workflows, and REST API integrations; 923 commits across 7 production deployments.
- Engineered agentic AI workflows (Claude, GPT-4o, Gemini) via MCP with provider-agnostic interfaces; built document processing (PDF gen, email parsing, S3) and full Playwright test coverage.

Graduate Research Assistant **NC State University** **April 2025 – Present**

- Deployed cloud-native genomic data platform on GCP (Cloud Run) with JBrowse 2 genome browser, enabling interactive exploration of 500GB+ strawberry and grape datasets for 100+ researchers worldwide.
- Improved ML model accuracy by 12% by applying hyperparameter tuning and time-series feature engineering on biological sensor data collected from genomic field experiments.

Software Developer Intern **Mphasis (HP Company)** **Jun 2024 – Aug 2024**

- Built 5+ AI workflows including health claim automation (OCR, NLP, ML); improved document processing accuracy by 25%.
- Automated AWS and GCP deployments via CI/CD; integrated Grafana monitoring and led model red-teaming for reliability.

AI/ML Intern **Capgemini** **Jun 2023 – Aug 2023**

- Led SDLC for data-driven ticket analysis system using NLP classification; reduced average resolution time by 15%.
- Built Kafka + Spark + Hadoop pipelines for batch and real-time data ingestion; cut ticket-processing latency by 20%.

Education

Raleigh, NC **North Carolina State University** **Aug 2024 – May 2026**

- Master of Science in Computer Science
- Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.

Mumbai, India **Mumbai University** **Aug 2020 – May 2024**

- Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.

Projects

Glance AI **Privacy-First Healthcare RAG** **2025**

- Designed per-tenant RAG pipeline for healthcare with strict vector isolation preventing cross-tenant data leakage and LLM hallucinations.
- Implemented deterministic retrieval for regulated environments; eliminated context bleed between tenants for factual LLM outputs.

AlgoPulse **Agentic AI Platform** **2025**

- Built agentic AI orchestration platform on Temporal.io and Kafka supporting multi-step reasoning pipelines; sustained 2,000 async inference requests per second under sustained production load.
- Replaced API polling with Supabase Edge Functions for real-time AI streaming to React Native clients; eliminated state loss under peak traffic.

PerCent **GenAI Finance App – HackNC24 Top 10** **2024**

- Built GenAI finance app with goal tracking, expense logging, and AI chatbot in React Native; ranked Top 10 of 100+ teams in 48-hour sprint.
- Designed RAG-based chatbot using Selenium for real-time financial data retrieval, Hugging Face embeddings, and Qwen-14B-Chat inference; reduced manual financial research time by 30% per user session.

Languages and Technologies

- **Languages:** Python, TypeScript, JavaScript, Java, SQL, Bash
- **AI/ML & Agentic:** Claude/GPT-4o/Gemini APIs, MCP, LangChain, RAG, Temporal.io, Hugging Face, prompt engineering
- **Frontend & Backend:** React, React Native, Node.js, Express, PostgreSQL, Redis, REST APIs, Auth0
- **Cloud & DevOps:** AWS, GCP, Docker, Terraform, GitHub Actions, Grafana, Sentry, Playwright

Additional Experience and Awards

- **IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- **National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons spanning ML, systems, and developer tools; built 3 complete MVPs end-to-end within 48-hour sprints, each with a working demo.
- **Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.
- **Technical Writing:** Published on LLM model collapse, AI agent security, audio AI at kulkarniaditya.com/blog